EXP NO-1

write a C program to simulate a deterministic finite Automata for the given language representing strings that start with a and end with a

C PROGRAM

```
#include <stdio.h>

#include <string.h>

int main()

{

  char str[100];

  int state = 0; // Initial state


  printf("Enter the string: ");

  scanf("%s", str);


  for (int i = 0; i < strlen(str); i++)

  {

    switch (state)

    {

      case 0: // Start state

        if (str[i] == 'a')

          state = 1;

        else

          state = 3; // Dead state

        break;


      case 1: // Starts with a and currently ends with a

        if (str[i] == 'a')

          state = 1;
```

```
            else if (str[i] == 'b')
                state = 2;
            break;

        case 2: // Starts with a and currently ends with b
            if (str[i] == 'a')
                state = 1;
            else if (str[i] == 'b')
                state = 2;
            break;

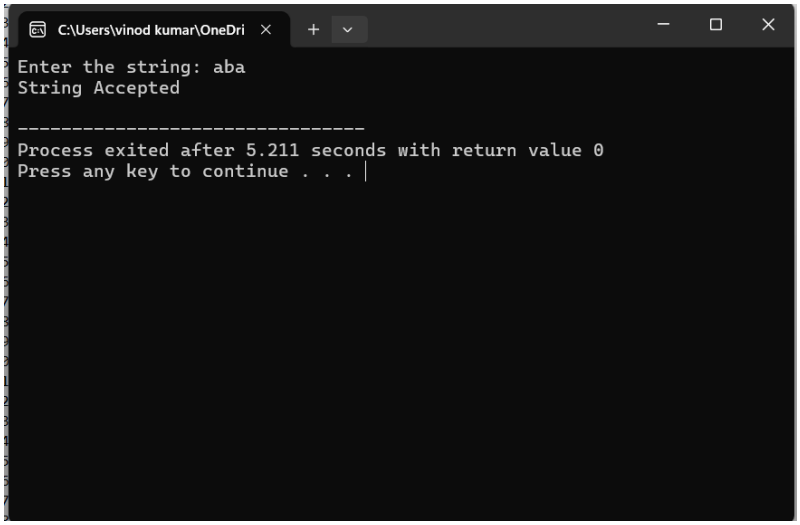
        case 3: // Dead state
            state = 3;
            break;
      }
   }

   if (state == 1)
      printf("String Accepted\n");
   else
      printf("String Rejected\n");

   return 0;
}
```

INPUT 1:-

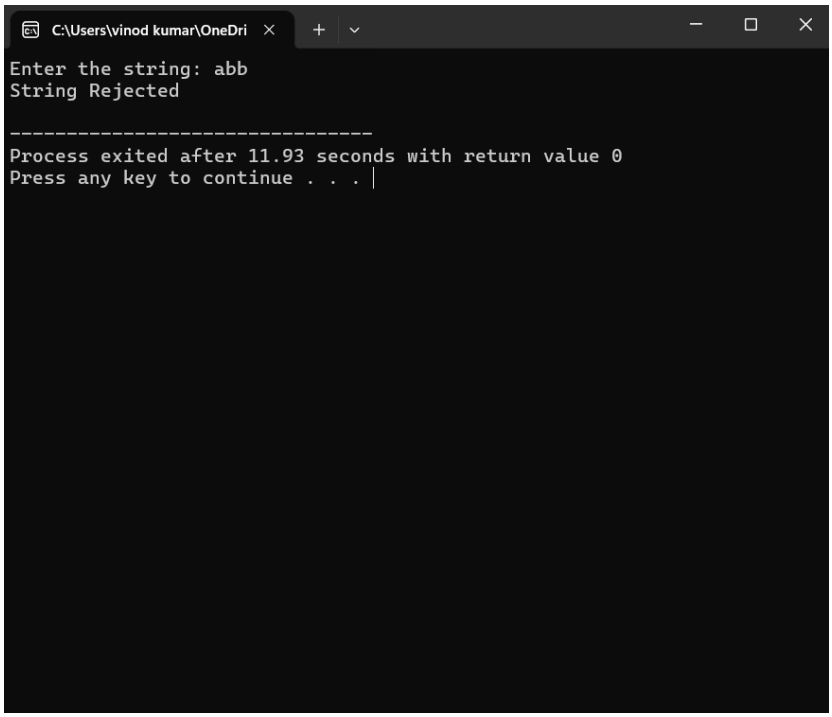
aba


C:\Users\vinod kumar\OneDri
Enter the string: aba
String Accepted

Process exited after 5.211 seconds with return value 0
Press any key to continue . . .

Input 2:-

abb


C:\Users\vinod kumar\OneDri
Enter the string: abb
String Rejected

Process exited after 11.93 seconds with return value 0
Press any key to continue . . .